

MODULE-7:

Here is a summary of the important points covered in the AWS Academy Cloud Architecting Module 7 Student Guide:

AWS Physical Infrastructure

- **AWS cloud infrastructure resides in data centers, which are grouped into Availability Zones (AZs) and further grouped into Regions.** Data centers have redundant power, networking, and connectivity. They house thousands of servers organized into racks, with routers and switches for routing traffic.
- **AZs offer low-latency connectivity (single-digit milliseconds) within a Region.** An AZ code combines the Region code with a letter (e.g., us-east-1a in the Northern Virginia Region).
- **Regions have latency in the tens of milliseconds.**
- **Virtual networks (overlay networks) are built atop the physical AWS network, allowing programmatic definition of network exposure.** Virtual networks use software-defined components like switches, routers, and firewalls.
- **AWS accounts have dedicated network resources, including a default virtual private cloud (VPC) in each public Region.** Each VPC belongs to one Region and can span multiple AZs.

Amazon VPC

- **An Amazon VPC is a customizable, logically isolated virtual network within a Region.** It simulates a traditional data center network and controls traffic flow using private IP addresses defined by a CIDR block.
- **VPC sizing is crucial, as the maximum size is a /16 netmask (65,536 IP addresses), and the minimum is a /28 netmask (16 IP addresses).** Allocating sufficient IP addresses initially is recommended, as adding them later is complex.
- **A VPC can be IPv4-only or dual-stack (IPv4 and IPv6).** IPv6 offers a larger address space, potentially better performance, and can be important given the cost for public IPv4 addresses that went into effect early in 2024.
- **A VPC has a main route table with routing rules.** The main route table has a default route (destination: VPC CIDR block, target: local), making all resources within the VPC reachable.
- **Subnets are segments of the VPC's IP address range within a single AZ and use their own route tables.** They contain routing policies and isolate resources with similar routing requirements. Each subnet should have a custom route table.
- **Public subnets provide direct internet access via an internet gateway and require both private and public IP addresses for instances.** An internet gateway allows communication between the VPC and the internet.
- **Elastic IP addresses are static, public IPv4 addresses that can be associated with an instance, load balancer, or network interface.** They remain associated with your AWS

account, even when not attached to a resource, unlike public IP addresses. You are charged for using Elastic IPs in some circumstances.

- **Private subnets lack direct internet access.** They are suitable for resources that should not be publicly reachable.

NAT Devices

- **Network Address Translation (NAT) devices allow resources in private subnets to connect to the internet while remaining private.** NAT devices replace the private IP address of the requesting instance with their own public IP address for outgoing traffic and perform the reverse translation for incoming traffic.
- **AWS offers NAT gateways (managed) and NAT instances (EC2-based).** NAT gateways are recommended for better availability, bandwidth, and ease of administration.

Securing Network Resources

- **Security should be implemented in layers, including secure protocols (e.g., TLS, HTTPS), security groups, and network ACLs.**
- **Security groups operate at the instance or network interface level, acting like apartment doors that only allow the tenant with a key to enter.** They are stateful (return traffic is automatically allowed) and can span multiple AZs.
- **Network ACLs operate at the subnet level, similar to a building doorman who allows only building residents to enter.** They are stateless (return traffic needs explicit rules) and support allow and deny rules.
- **AWS Network Firewall provides an additional security layer for VPCs.** It's a stateful firewall and intrusion detection/prevention service that inspects traffic entering the VPC.
- **Bastion hosts in public subnets provide secure access to resources in private subnets, limiting direct external access.**

Connecting to Managed AWS Services

- **VPC endpoints offer private connectivity to AWS managed services.** Two types exist: interface VPC endpoints and gateway VPC endpoints.
- **Interface VPC endpoints use AWS PrivateLink for private connections to most AWS managed services.** They are secured using IAM resource policies.
- **Gateway VPC endpoints connect directly to Amazon S3 and Amazon DynamoDB using route tables, bypassing PrivateLink.** They offer unlimited bandwidth and no additional cost.
- **Gateway Load Balancer endpoints provide private connections between security appliances and applications across VPCs.** They allow for traffic inspection and security appliance integration.

Monitoring Your Network

- **VPC Flow Logs capture detailed information about network traffic in a VPC.** They can be delivered to Amazon CloudWatch Logs, Amazon S3, or Amazon Kinesis Data Firehose.
- **Flow logs require appropriate IAM permissions for user access.** Policies are used to grant or deny access to flow log actions (create, describe, delete).
- **Flow log records contain detailed information about traffic, including source/destination IP addresses, ports, protocols, timestamps, and actions (accept/reject).**
- **Additional VPC troubleshooting tools include Reachability Analyzer for connectivity testing, Network Access Analyzer for identifying unintended access, and Traffic Mirroring for security and monitoring purposes.**

AWS Well-Architected Framework

- **The AWS Well-Architected Framework provides best practices for designing, operating, and maintaining workloads and networks.**
- **The Reliability pillar focuses on network resiliency through proper IP subnet allocation to accommodate growth and availability.**
- **The Security pillar emphasizes a Zero Trust approach, creating network layers, controlling traffic at all layers, and implementing inspection and protection mechanisms.**
- **The Performance Efficiency pillar addresses optimizing network architecture by understanding performance impacts, evaluating available networking features, and choosing appropriate protocols.**
- **The Cost Optimization pillar highlights selecting the most cost-effective AWS Regions and pricing models for network resources.**

Lab Activities

The student guide also includes details about guided and challenge labs focused on creating and configuring VPCs, enhancing network security, and applying Well-Architected Framework principles.

Sample Exam Question

The document provides a sample exam question and answer explanation, emphasizing the importance of understanding NAT gateways and route table configurations for enabling internet access from private subnets.